

IIT Jodhpur Qualifier - BS

QUALIFIER EXAMINATION | Time: 2 Hours | Maximum Marks: 198

Question Distribution

- **VARC:** 24 questions (suggested time 40 min)
- **DILR:** 18 questions (suggested time 40 min)
- **QA:** 24 questions (suggested time 40 min)
- Total: 66 questions | MCQ 60 | TITA (non-MCQ) 6

Marking Scheme

- MCQ: +3 for a correct answer, -1 for a wrong answer.
- TITA / Non-MCQ: +3 for a correct answer, no negative marking.
- Unanswered questions: no marks awarded or deducted.

General Instructions

- Read each question carefully before attempting.
- For MCQs, choose the single most appropriate option (A-D, or A-E where shown).
- For TITA questions, type the final numerical answer.
- Total duration is 2 hours; each section has a suggested 40-minute window.
- Sections may be attempted in any order.

This is the answer-marked verification copy (correct option in green, with solution) and mirrors the content of the upload CSV.

Section: VARC (suggested 40 min)

Instructions (VARC-RC1): study the shared passage below and answer its 4 question(s).

For most of human history, the night sky was a shared inheritance: a band of stars visible to anyone who looked upward. That inheritance is quietly dissolving. The spread of electrified cities has scattered artificial light into the atmosphere, where dust and moisture scatter it further, producing a diffuse glow that astronomers call skyglow. Under this veil, the faint light of distant stars is simply drowned out. Surveys now suggest that a large share of people alive today have never seen the Milky Way from where they live, and that the brightness of the night sky over inhabited regions is rising by a few per cent each year.

The consequences reach well beyond stargazing. Many organisms time their behaviour by natural cycles of darkness. Migrating birds, which navigate partly by the stars, can be disoriented by illuminated buildings; insects exhaust themselves circling lamps; hatchling sea turtles, which instinctively crawl toward the brightest horizon, may turn inland toward roads instead of out to sea. In each case a cue that was reliable for millions of years has been overwritten by a human signal.

What makes light pollution unusual among environmental problems is how readily it can be reversed. Carbon dioxide lingers in the atmosphere for centuries; a poorly aimed streetlamp stops polluting the moment it is switched off or shielded. Lighting that points downward, that is no brighter than a task requires, and that is dimmed when no one is present can preserve both safety and darkness. The obstacle is not technical but a habit of mind that equates more light with more progress. Recovering the night sky, then, asks less for new invention than for a willingness to use, more thoughtfully, the light we already command.

Q1. [VARC-RC1 · mcsc · diff 0.5]

Which of the following best expresses the main idea of the passage?

- A) Artificial light has now made it completely impossible for most people on Earth to see any stars at all from any of the places where they currently live and work.
- B) The disorientation of migrating birds, circling insects, and stranded sea turtles is now the single most urgent environmental crisis facing the entire planet.
- C) Light pollution is a serious but unusually reversible problem, and addressing it depends more on changing our lighting habits than on inventing new technology. ✓
- D) Astronomers, rather than elected officials, should be given final authority over urban lighting decisions because they understand skyglow better than city planners.

Answer: C) Light pollution is a serious but unusually reversible problem, and addressing it depends more on changing our lighting habits than on inventing new technology.

Solution: The key captures both halves of the passage: light pollution is harmful yet uniquely reversible, and the fix is mainly a matter of changing habits rather than inventing new technology. The 'cannot see any stars at all' choice overstates the claim, since the passage says many cannot see the Milky Way, not all stars. The 'wildlife disorientation is the single most urgent crisis' choice elevates one example into the central thesis, which the passage never does. The 'astronomers should have final authority' choice introduces a policy recommendation the passage does not make.

Q2. [VARC-RC1 · mcsc · diff 0.5]

According to the passage, the author makes all of the following claims EXCEPT:

- A) Skyglow is produced partly because dust and moisture in the atmosphere scatter artificial light.
- B) Reducing light pollution requires expensive new lighting technologies that have not yet been invented. ✓
- C) Some animals rely on natural darkness or starlight as a behavioural cue that artificial light can disrupt.
- D) The brightness of the night sky over inhabited regions has been increasing year by year.

Answer: B) Reducing light pollution requires expensive new lighting technologies that have not yet been invented.

Solution: The claim that reducing light pollution needs expensive uninvented technology is the EXCEPT answer, since the passage argues the opposite: the fix needs not new invention but more thoughtful use of light already available. The statement about dust and moisture scattering light restates the stated mechanism of skyglow. The claim that some animals rely on natural darkness paraphrases the birds, insects and turtles examples. The statement that night-sky brightness has been rising directly matches the claim that it grows a few per cent each year.

Q3. [VARC-RC1 · mcsc · diff 0.5]

Why does the author contrast light pollution with carbon dioxide?

- A) To argue that carbon dioxide is, in the end, a far less serious threat to migrating wildlife than artificial light is.
- B) To suggest that the two problems are caused by exactly the same kind of polluting industrial activity in modern cities.
- C) To prove that most environmental problems generally cannot be solved at all without first achieving major scientific breakthroughs.
- D) To highlight that light pollution can be reversed almost at once, whereas carbon dioxide lingers for a very long time. ✓

Answer: D) To highlight that light pollution can be reversed almost at once, whereas carbon dioxide lingers for a very long time.

Solution: The 'reversed almost at once while carbon dioxide lingers' choice matches the explicit contrast: CO₂ lingers for centuries while a lamp stops polluting the instant it is shielded or switched off, illustrating reversibility. The 'carbon dioxide is less serious for wildlife' choice invents a ranking of threats not made in the text. The 'same polluting industrial activity' choice misreads the contrast as a shared cause. The 'breakthroughs needed to solve problems' choice contradicts the passage, which says the fix needs willingness, not breakthroughs.

Q4. [VARC-RC1 · mcsc · diff 0.5]

It can be inferred that the author would most likely agree with which of the following statements?

- A) A community could cut its light pollution without sacrificing safety by aiming lamps downward and dimming them when no one is around. ✓
- B) The loss of the visible Milky Way is, regrettably, an entirely unavoidable cost of building any modern, fully electrified society.
- C) Cities ought to abandon their street lighting almost entirely in order to recover the lost visibility of the dark and starry night sky.
- D) Concern about the fading night sky is mainly a private preoccupation of professional astronomers rather than a truly broad public issue.

Answer: A) A community could cut its light pollution without sacrificing safety by aiming lamps downward and dimming them when no one is around.

Solution: The 'cut light pollution without sacrificing safety' choice follows directly from the claim that downward-pointing, appropriately bright, motion-dimmed lighting can preserve both safety and darkness. The 'unavoidable cost of an electrified society' choice is too extreme, since the passage seeks thoughtful use, not abandonment, of light. The 'abandon street lighting almost entirely' choice contradicts the framing of the sky as a shared inheritance and the wildlife effects. The 'mainly a preoccupation of astronomers' choice conflicts with the central point that the loss is reversible and broadly relevant.

Instructions (VARC-RC2): study the shared passage below and answer its 4 question(s).

When a result in science is described as 'replicated,' it means that an independent team, following the same procedure, has obtained a similar finding. Replication is often called the gatekeeper of reliable knowledge, because a single study, however carefully done, can be misled by chance, by an unusual sample, or by an undetected flaw in its design. Only when a result reappears under different hands does the community gain confidence that it reflects something real rather than an accident of one laboratory.

In recent years, however, attempts to repeat well-known studies have frequently failed to reproduce the original effects. Several factors contribute. Researchers face strong incentives to publish striking, novel results, while careful repetitions of someone else's work attract little reward and are seldom published. Small samples can yield impressive-looking outcomes that vanish when the study is enlarged. And the freedom to analyse data in many ways allows a researcher, even unintentionally, to keep adjusting the analysis until a pattern emerges.

None of this means that science is broken. The very fact that the failures are being documented is itself a product of the scientific method turning its scrutiny on its own practices. Reforms are already spreading: some journals now invite researchers to register their hypotheses and planned analyses before collecting data, and commit to publishing the results whether or not they are striking. Others reserve space specifically for replication studies. What looks at first like a crisis of confidence may be better understood as a discipline learning, in public, to hold itself to its own standards.

Q5. [VARC-RC2 · mcsc · diff 0.5]

Which of the following best states the central argument of the passage?

- A) Scientific journals are the primary cause of unreliable published findings and therefore should no longer be trusted at all by readers.
- B) Although many well-known studies have failed to replicate, this reflects science correcting itself rather than a breakdown of the method. ✓
- C) Small samples are, on their own, the single underlying reason that so many famous studies have proven difficult to reproduce reliably.
- D) Replication is essentially impossible in actual practice because no two separate laboratories ever follow exactly the same detailed procedure.

Answer: B) Although many well-known studies have failed to replicate, this reflects science correcting itself rather than a breakdown of the method.

Solution: The 'science correcting itself rather than a breakdown' choice captures the thesis: replication failures are real but signal a self-correcting discipline, not a broken one, supported by the closing paragraph. The 'journals are the primary cause and should not be trusted' choice overstates the role of journals, which the passage treats as one factor and a source of reform. The 'small samples are the single reason' choice isolates one of several listed causes as the only one. The 'replication is essentially impossible' choice contradicts the definition of replication offered in the first paragraph.

Q6. [VARC-RC2 · mcsc · diff 0.5]

The author identifies all of the following as factors contributing to replication failures EXCEPT:

- A) The freedom to analyse data in many ways until a pattern happens to appear.
- B) A deliberate effort by most researchers to fabricate data in order to deceive their peers. ✓
- C) Incentives that reward novel findings while offering little reward for repeating others' work.
- D) Small samples that can produce striking results which disappear when the study is enlarged.

Answer: B) A deliberate effort by most researchers to fabricate data in order to deceive their peers.

Solution: The 'deliberate effort to fabricate data' choice is the EXCEPT answer: the passage attributes problems to incentives, sample size, and analytic flexibility, and explicitly notes that adjusting analyses can happen 'even unintentionally' rather than via deliberate fraud. The 'freedom to analyse data many ways', 'incentives rewarding novel findings', and 'small samples that vanish when enlarged' choices each restate one of the three factors the passage names directly.

Q7. [VARC-RC2 · mcsc · diff 0.5]

What is the main purpose of registering hypotheses and planned analyses before collecting data, as described in the passage?

- A) To guarantee that each and every registered study will reliably end up producing some kind of novel and publishable finding.
- B) To replace careful replication studies entirely, so that any repetition of other people's earlier work is no longer needed at all.
- C) To allow researchers to choose freely, after they have seen the data, which analysis presents their results most favourably.
- D) To limit the freedom to adjust the analysis until a desired pattern appears, and to ensure results are reported either way. ✓

Answer: D) To limit the freedom to adjust the analysis until a desired pattern appears, and to ensure results are reported either way.

Solution: The 'limit the freedom to adjust analysis and report either way' choice follows from the passage: pre-registration commits researchers to a plan and to publishing results whether striking or not, curbing the analytic flexibility named earlier. The 'guarantee a publishable novel finding' choice misstates the goal as guaranteeing novelty. The 'replace replication studies entirely' choice wrongly claims it replaces replication, which the passage lists as a separate reform. The 'choose freely after seeing the data' choice describes the very practice pre-registration is meant to prevent.

Q8. [VARC-RC2 · mcsc · diff 0.5]

It can be inferred from the passage that the author regards the documentation of replication failures as:

- A) definite proof that most published scientific results will, sooner or later, be shown to be entirely false.
- B) an unfortunate and costly distraction that needlessly diverts attention away from producing genuinely novel discoveries.
- C) a serious problem that can only ever be solved by abandoning the basic requirement of replication altogether.
- D) evidence that science is applying its own critical methods to itself rather than a sign of fundamental decay. ✓

Answer: D) evidence that science is applying its own critical methods to itself rather than a sign of fundamental decay.

Solution: The 'science applying its own critical methods to itself' choice reflects the statement that documenting failures is itself the scientific method scrutinising its own practices, and the reframing of the 'crisis' as healthy self-correction. The 'proof that most published results are false' choice overgeneralises to most results being false, which the text does not claim. The 'unfortunate distraction from novel discoveries' choice contradicts the positive framing of self-scrutiny. The 'solved only by abandoning replication' choice reverses the argument, which treats replication as the gatekeeper to be strengthened, not abandoned.

Instructions (VARC-RC3): study the shared passage below and answer its 4 question(s).

A city map is never a perfect copy of the city it represents. If it were, it would have to be exactly as large and as complicated as the city itself, and so would be useless to a traveller. The value of a map lies precisely in what it leaves out. By dropping countless details and keeping only roads, rivers, and a few landmarks, a map becomes small enough to hold in the hand and clear enough to guide a journey.

The same principle applies to scientific models. A model of the weather, or of a population of animals, does not try to capture every raindrop or every individual creature. Instead it keeps the features that matter for the question being asked and ignores the rest. A model built to predict tomorrow's temperature can safely leave out the colour of people's umbrellas; a model of how a disease spreads can ignore the brand of car that infected people drive. Choosing what to leave out is not a weakness of a model but the very thing that makes it useful.

This has a practical lesson. When we judge a model, the right question is not 'Is it complete?' but 'Is it useful for its purpose?' A simple model that answers the question at hand may be far more valuable than an elaborate one that tries to include everything and so becomes impossible to understand. As one saying among scientists puts it, all models are wrong, but some are useful. Wisdom lies in matching the simplicity of the model to the job it must do.

Q9. [VARC-RC3 · mcsc · diff 0]

Which of the following best captures the main idea of the passage?

- A) Scientific models should always try to include as much fine-grained detail as they possibly can in order to be truly accurate.
- B) Maps are, in the end, far more reliable guides to a city than scientific models can ever be to the natural world.
- C) A good model, much like a good map, is valuable precisely because it leaves out details that do not matter for its purpose. ✓
- D) The colour of people's umbrellas and the brand of a car are in fact important variables in both weather and disease models.

Answer: C) A good model, much like a good map, is valuable precisely because it leaves out details that do not matter for its purpose.

Solution: The 'valuable because it leaves out details that do not matter' choice states the central analogy: usefulness comes from selective omission, true of both maps and models. The 'include as much detail as possible' choice contradicts the passage, which praises leaving things out. The 'maps are more reliable than models' choice invents a comparison of reliability the passage never makes. The 'umbrella colour and car brand are important variables' choice reverses the examples, which are given precisely as details that can safely be ignored.

Q10. [VARC-RC3 · mcsc · diff 0]

Why does the author say that a perfectly complete map would be useless?

- A) Because maps are always rather less accurate than the kind of scientific models that are discussed later on in the passage.
- B) Because it would have to be as large and as complicated as the city itself, and so could not guide a traveller. ✓
- C) Because real cities change far too quickly for any printed map of them to remain fully correct for very long.
- D) Because no ordinary traveller is ever genuinely interested in the roads, the rivers, or the landmarks of a city.

Answer: B) Because it would have to be as large and as complicated as the city itself, and so could not guide a traveller.

Solution: The 'as large and complicated as the city itself' choice paraphrases the explicit reason: a complete map would match the city in size and complexity and so help no one. The 'maps less accurate than models' choice introduces an accuracy comparison not present in the text. The 'cities change too quickly' choice raises the rate of urban change, which the passage never mentions. The 'travellers not interested in roads or rivers' choice contradicts the passage, which says maps keep roads, rivers, and landmarks because they matter.

Q11. [VARC-RC3 · mcsc · diff 0.5]

It can be inferred that the author would most likely judge a scientific model by asking:

- A) whether it serves its intended purpose well, even if it leaves out many features of the real situation. ✓
- B) whether it manages to include every single variable that could conceivably affect the final outcome in any way.
- C) whether it is noticeably more detailed and elaborate than any competing model that addresses the same question.
- D) whether it perfectly reproduces the whole of reality without leaving out or omitting anything at all whatsoever.

Answer: A) whether it serves its intended purpose well, even if it leaves out many features of the real situation.

Solution: The 'serves its intended purpose well even if it omits features' choice follows from the stated criterion 'Is it useful for its purpose?' and the praise of simple, fit-for-purpose models. The 'includes every variable that could affect the outcome' choice contradicts the argument against trying to include everything. The 'more detailed than any competing model' choice equates value with maximum detail, which the passage rejects. The 'perfectly reproduces all of reality' choice describes the impossible complete model the passage dismisses as useless.

Q12. [VARC-RC3 · mcsc · diff 0.5]

What is the main function of the saying 'all models are wrong, but some are useful' in the passage?

- A) To warn readers firmly that scientific models really should not be trusted when making any practical decision.
- B) To introduce a fresh objection that the author then goes on to refute carefully and in some detail.
- C) To claim that maps, quite unlike scientific models, are essentially never wrong about the cities they represent.

D) To reinforce the author's point that a model need not be complete in order to be valuable for its purpose. ✓

Answer: D) To reinforce the author's point that a model need not be complete in order to be valuable for its purpose.

Solution: The 'reinforce that a model need not be complete to be valuable' choice fits the saying's placement: it summarises and reinforces the thesis that usefulness, not completeness, is the right standard. The 'warn that models should not be trusted' choice misreads the saying as a blanket distrust of models. The 'introduce an objection the author refutes' choice wrongly treats it as something the author rebuts, when it supports the argument. The 'claim maps are never wrong' choice invents a contrast in which maps are infallible, contradicting the opening that no map is a perfect copy.

Q13. [VARC-ODD-1 · mcsc · diff 0.5]

Four of the following five sentences, when arranged properly, form a coherent paragraph on a single theme; one sentence does not belong.

- (1) Coral reefs occupy less than one percent of the ocean floor, yet they shelter roughly a quarter of all marine species.
- (2) The dense limestone structures act as nurseries where countless fish spawn and mature.
- (3) They also buffer nearby coastlines by absorbing much of the energy carried by incoming waves.
- (4) Many tropical resorts now offer guided diving tours that bring tourists close to these formations.
- (5) Because of this ecological richness, the loss of a single reef can ripple through an entire marine food web.

Which sentence is the odd one out?

- A) 3
- B) 5
- C) 4 ✓
- D) 1
- E) 2

Answer: C) 4

Solution: Sentences 1, 2, 3 and 5 all describe the ecological importance of coral reefs (biodiversity, nursery role, coastal protection, food-web impact). Sentence 4 shifts to tourism and resort marketing, an unrelated commercial angle, so it is the misfit. The other numbers are wrong because each of those sentences contributes to the single ecological theme.

Q14. [VARC-ODD-2 · mcsc · diff 0.5]

Four of the following five sentences, when arranged properly, form a coherent paragraph on a single theme; one sentence does not belong.

- (1) The printing press, introduced in fifteenth-century Europe, dramatically lowered the cost of reproducing texts.
- (2) Books that had once taken months to copy by hand could now be produced in days.
- (3) As prices fell, reading spread beyond monasteries and universities to ordinary townspeople.
- (4) This wider circulation of ideas helped fuel movements in science and religion.
- (5) Modern libraries increasingly lend electronic books that readers download to personal devices.

Which sentence is the odd one out?

- A) 5 ✓
- B) 1
- C) 2
- D) 3
- E) 4

Answer: A) 5

Solution: Sentences 1 to 4 trace a single historical narrative about the printing press and its consequences for the spread of reading and ideas. Sentence 5 jumps to present-day e-book lending, a different era and technology, breaking the chain, so it is the odd one out. Each of the other sentences advances the printing-press story.

Q15. [VARC-ODD-3 · mcsc · diff 0.5]

Four of the following five sentences, when arranged properly, form a coherent paragraph on a single theme; one sentence does not belong.

- (1) Honeybees communicate the location of food through a movement known as the waggle dance.
- (2) Beekeepers harvest surplus honey by carefully removing frames from the hive.
- (3) A returning forager runs in a figure-eight pattern on the vertical comb.
- (4) The angle of the run relative to gravity encodes the direction of the food source.
- (5) The duration of the waggle phase signals how far away the food lies.

Which sentence is the odd one out?

- A) 5
- B) 2 ✓
- C) 1
- D) 4
- E) 3

Answer: B) 2

Solution: Sentences 1, 3, 4 and 5 form a tight explanation of how the honeybee waggle dance conveys both the direction and distance of food. Sentence 2 turns to human honey harvesting, an unrelated beekeeping activity, so it does not belong. The remaining sentences each describe a feature of the communicative dance.

Q16. [VARC-JUMBLE-1 · mcsc · diff 0.5]

The following four sentences, labelled 1 to 4, can be arranged to form a coherent paragraph.

1. The result was a network of stone roads stretching across the whole empire.
2. Ancient Rome needed a fast way to move its armies between distant provinces.
3. To meet this need, engineers laid down durable paved routes wherever the legions marched.
4. Centuries later, many of these same roads still guide the paths of modern highways.

Select the correct sequence of sentences.

- A) 2341
- B) 3214
- C) 2314 ✓
- D) 2134

Answer: C) 2314

Solution: Sentence 2 states the problem (Rome needed fast troop movement), 3 gives the engineers' response, 1 reports the result (a road network), and 4 closes with the long-term legacy. The order 2-3-1-4 is the only sequence in which cause, action, outcome and aftermath follow naturally; the distractors break this problem-solution-result chain.

Q17. [VARC-JUMBLE-2 · mcsc · diff 0.5]

The following four sentences, labelled 1 to 4, can be arranged to form a coherent paragraph.

1. Over time, this trapped heat raises the planet's average surface temperature.
2. Certain gases in the atmosphere allow sunlight to pass through but absorb the heat that the ground radiates back.
3. This natural process is what scientists call the greenhouse effect.
4. When the concentration of these gases rises, more heat is retained than before.

Select the correct sequence of sentences.

- A) 2341
- B) 3241
- C) 2314 ✓
- D) 2431

Answer: C) 2314

Solution: Sentence 2 introduces the gases and what they do, 3 names the process just described (the greenhouse effect), 1 states the gradual consequence of trapped heat, and 4 adds the amplifying condition when gas levels rise. The 2-3-1-4 order moves from mechanism to label to effect to intensification; the alternatives misplace the naming sentence or the consequence.

Q18. [VARC-JUMBLE-3 · mcsc · diff 0.5]

The following four sentences, labelled 1 to 4, can be arranged to form a coherent paragraph.

1. She began by sketching rough thumbnails to test different compositions.
2. Maria had been asked to design the cover for the school magazine.
3. Once a layout felt right, she refined it into a detailed digital drawing.
4. Finally, she added colour and sent the finished file to the printer.

Select the correct sequence of sentences.

- A) 1234
- B) 2143
- C) 2134 ✓
- D) 2314

Answer: C) 2134

Solution: Sentence 2 sets up the task (Maria asked to design a cover), 1 starts the work with thumbnails, 3 refines a chosen layout, and 4 completes the process with colour and printing. The 2-1-3-4 sequence follows the natural order of a design workflow from assignment to delivery; the distractors either open mid-process or reverse the refining and finishing steps.

Q19. [VARC-JUMBLE-4 · mcsc · diff 0.5]

The following four sentences, labelled 1 to 4, can be arranged to form a coherent paragraph.

1. At first the seedlings were too fragile to survive outdoors.
2. The gardeners started the tomato plants indoors during early spring.
3. So they kept the young plants under warm lamps for several weeks.
4. Only when the stems had thickened did the gardeners move them into the open beds.

Select the correct sequence of sentences.

- A) 2314
- B) 1234
- C) 2134 ✓
- D) 2143

Answer: C) 2134

Solution: Sentence 2 establishes that the plants were started indoors, 1 explains why they could not yet go outside (too fragile), 3 gives the response (kept under lamps), and 4 marks the transfer outdoors once strong enough. The 2-1-3-4 order links the indoor start, the problem, the remedy and the eventual move; the distractors break this cause-and-effect progression.

Q20. [VARC-SUMMARY-1 · mcsc · diff 0]

Public libraries today do far more than lend books. Many now offer free internet access, run job-search workshops, host story hours for children, and lend tools or musical instruments. For people without these resources at home, the library has become a community centre as much as a place to read.

Which of the following best summarises the paragraph?

- A) Modern public libraries have grown into broad community centres that now offer many services going well beyond simply lending out books. ✓
- B) Public libraries today mainly function as places that lend out musical instruments and household tools to the local residents who ask.
- C) Libraries have lost all interest in books and now devote themselves entirely to providing free internet access to their local visitors.
- D) The single most valuable thing libraries do now is to host the enjoyable weekly story hours for the young children of the community.

Answer: A) Modern public libraries have grown into broad community centres that now offer many services going well beyond simply lending out books.

Solution: The 'grown into broad community centres beyond lending books' choice captures the paragraph's main point: libraries now serve as community centres providing many services in addition to books. The 'mainly lend instruments and tools' choice is too narrow, naming only one detail. The 'lost all interest in books for internet access' choice distorts the text, which says libraries do more than lend books, not that they abandoned books. The 'story hours are the single most valuable thing' choice is a minor detail, not the overall idea.

Q21. [VARC-SUMMARY-2 · mcsc · diff 0.5]

Sleep researchers have found that pulling an all-nighter before an exam usually backfires. While extra waking hours seem to allow more study, the tired brain struggles to store new information, and fatigue slows recall the next day. Students who sleep a full night generally remember more, even though they spent fewer hours reviewing.

Which of the following best summarises the paragraph?

- A) Students who choose to stay awake all night before an exam manage to study far more material and so tend to recall it better.
- B) Sacrificing sleep to study longer tends to weaken memory and recall, so a full night of proper rest usually serves students better. ✓
- C) How much a student sleeps has no measurable effect at all on how well that student later performs on a difficult written examination.
- D) Sleep researchers now firmly advise students that they should never review any of their own notes during the evening before an exam.

Answer: B) Sacrificing sleep to study longer tends to weaken memory and recall, so a full night of proper rest usually serves students better.

Solution: The 'sacrificing sleep weakens memory so rest serves better' choice conveys the central trade-off: the extra study time gained by skipping sleep is outweighed by weaker memory, so resting is generally better. The 'staying awake lets you recall more' choice repeats only the surface claim the passage rebuts. The 'sleep has no measurable effect' choice contradicts the finding that sleep helps recall. The 'never review notes the evening before' choice overreaches into advice the paragraph never gives.

Q22. [VARC-SUMMARY-3 · mcsc · diff 0.5]

Electric cars produce no exhaust as they drive, which clearly improves air quality in cities. Yet whether they are truly cleaner depends on how their electricity is generated. Where power comes mostly from coal, the overall emissions can rival those of petrol cars; where it comes from renewables, the advantage is large.

Which of the following best summarises the paragraph?

- A) Coal-fired power stations still remain the main source of almost all of the electricity that is used to charge most electric cars today.
- B) Electric cars are always far cleaner than ordinary petrol cars in every situation, no matter how their electrical power happens to be generated.
- C) Electric cars release no pollution at all from their tailpipes while they are being driven along busy city streets and the open roads beyond.
- D) Electric cars cut local pollution, but their full environmental benefit ultimately depends on how the electricity charging them is produced. ✓

Answer: D) Electric cars cut local pollution, but their full environmental benefit ultimately depends on how the electricity charging them is produced.

Solution: The 'cut local pollution but full benefit depends on how electricity is produced' choice balances both halves of the paragraph: a clear local benefit, but overall cleanliness depends on the electricity source. The 'no tailpipe pollution while driving' choice states only one detail and misses the central condition. The 'always far cleaner in every situation' choice asserts a fact the passage does not claim. The 'coal stations supply almost all charging electricity' choice overreaches by treating the coal caveat as a universal fact.

Q23. [VARC-SUMMARY-4 · mcsc · diff 0.5]

Volunteering is often described as pure generosity, but studies suggest the giver gains too. People who regularly help others report lower stress, a stronger sense of purpose, and wider social networks. In this sense, the act of giving time quietly returns a benefit to the volunteer.

Which of the following best summarises the paragraph?

A) Although volunteering clearly helps other people in many ways, research indicates that it also brings the volunteer real personal benefits. ✓

B) Volunteering is best understood as a wholly selfless act of giving that quietly returns no benefit of any real kind to the person doing it.

C) People generally choose to volunteer mainly because doing so lets them expand and strengthen their own valuable personal social networks.

D) Regular volunteers consistently report that they now feel considerably less stressed and rather more relaxed than they did before they began.

Answer: A) Although volunteering clearly helps other people in many ways, research indicates that it also brings the volunteer real personal benefits.

Solution: The 'helps others yet also brings the volunteer real personal benefits' choice captures the paragraph's twist: volunteering helps others yet also rewards the giver. The 'wholly selfless act returning no benefit' choice contradicts the passage, which argues the giver does gain. The 'volunteer mainly to expand social networks' choice narrows the motive to one benefit and misstates intention. The 'feel less stressed and more relaxed' choice reports a single detail rather than the overall claim that giving returns several benefits.

Q24. [VARC-SUMMARY-5 · mcsc · diff 0.5]

When a new word enters a language, it rarely arrives by official decree. Instead, speakers begin using it, others copy them, and dictionaries record the word only after it has already taken hold. Language, in other words, is shaped from the bottom up by ordinary use rather than handed down by authorities.

Which of the following best summarises the paragraph?

A) Dictionaries hold the real authority here and are the bodies responsible for officially deciding which new words a language may properly adopt.

B) Most new words manage to enter a language only after some official government body has formally reviewed and then finally approved each one.

C) New words spread through everyday use among ordinary speakers, and dictionaries follow rather than simply lead this gradual change over time. ✓

D) Speakers occasionally invent striking new words that many other people across the wider language community then choose to copy and reuse.

Answer: C) New words spread through everyday use among ordinary speakers, and dictionaries follow rather than simply lead this gradual change over time.

Solution: The 'new words spread through everyday use while dictionaries follow' choice states the main idea: language changes from the bottom up through ordinary use, with dictionaries recording words after the fact. The 'dictionaries hold the real authority to decide' choice reverses this by giving dictionaries the deciding role. The 'words enter only after government approval' choice captures only a fragment of the process and omits the role of dictionaries. The 'speakers occasionally invent striking words others copy' choice contradicts the passage's denial of official decree by ignoring the dictionaries-follow point.

Section: DILR (suggested 40 min)

Instructions (DILR-1): study the shared data below and answer its 3 question(s).

Four teams, Alfa, Bravo, Comet and Delta, play a single round-robin tournament: every team plays every other team exactly once (6 matches in all). A win is worth 3 points, a tie is worth 1 point to each team, and a loss is worth 0 points. The following facts are known about the completed tournament:

- Alfa won every match it played. - Delta lost every match it played. - Exactly one match in the whole tournament ended in a tie. - That single tie was the match between Bravo and Comet.

Q25. [DILR-1 · mcsc · diff 0.5]

How many points did Comet earn in the tournament?

- A) 4 points ✓
- B) 7 points
- C) 3 points
- D) 6 points

Answer: A) 4 points

Solution: Alfa won all 3 of its matches, so Alfa beat Bravo, Comet and Delta. Delta lost all 3. The only tie was Bravo vs Comet, so every other match had a winner. Comet thus lost to Alfa (0), tied Bravo (1) and beat Delta (3): total $1 + 3 = 4$ points. Distractor 3 = only the win; distractor 6 = double-counting the tie as 3+3; distractor 7 = treating the Bravo match as a win (3+3+1).

Q26. [DILR-1 · mcsc · diff 0.5]

How many of the 6 matches ended with a winner (that is, did NOT end in a tie)?

- A) 6 matches
- B) 4 matches
- C) 5 matches ✓
- D) 3 matches

Answer: C) 5 matches

Solution: There are 6 matches in a round-robin among 4 teams. Exactly one match (Bravo vs Comet) ended in a tie, so the remaining $6 - 1 = 5$ matches ended with a winner. Distractor 6 ignores the tie; distractors 4 and 3 over-subtract.

Q27. [DILR-1 · mcsc · diff 0.5]

Which statement about the final points table is correct?

- A) Comet finished ahead of Bravo on points, with Alfa on top.
- B) Bravo and Comet finished level on points, behind Alfa and ahead of Delta. ✓
- C) Delta finished level on points with Bravo, behind Alfa and Comet.
- D) Alfa and Bravo finished level on points at the top, ahead of Comet and Delta.

Answer: B) Bravo and Comet finished level on points, behind Alfa and ahead of Delta.

Solution: Final points: Alfa 9 (three wins), Delta 0 (three losses). Bravo = loss to Alfa (0) + tie with Comet (1) + win over Delta (3) = 4; Comet = loss to Alfa (0) + tie with Bravo (1) + win over Delta (3) = 4. So Bravo and Comet are level on 4, below Alfa's 9 and above Delta's 0. The claim that Comet finished ahead of Bravo is false (they are equal); the claim that Delta is level with Bravo is false (Delta has 0); the claim that Alfa and Bravo are level at the top is false (Alfa alone leads on 9).

Instructions (DILR-2): study the shared data below and answer its 3 question(s).

Three friends, Mira, Nina and Owen, each take the same short quiz of 5 questions. Every correct answer is worth 3 points; a wrong or blank answer scores 0 points. So each person's score equals 3 times the number of questions they answered correctly. The following facts are known:

- Between them, the three friends answered exactly 9 questions correctly. - Mira answered more questions correctly than Nina, and Nina answered more correctly than Owen. - Owen answered exactly 2 questions correctly.
- No one answered all 5 questions correctly.

Q28. [DILR-2 · mcsc · diff 0.5]

How many points did Mira score?

- A) 9 points
- B) 6 points
- C) 12 points ✓
- D) 15 points

Answer: C) 12 points

Solution: Owen got 2 correct. The three totals must sum to 9, with $\text{Mira} > \text{Nina} > \text{Owen} = 2$ and no one above 4 correct (since nobody got all 5). So $\text{Nina} = 3$ and $\text{Mira} = 4$ ($4 + 3 + 2 = 9$). Mira's score = $3 \times 4 = 12$. Distractor 9 = Nina's score; distractor 6 = Owen's score; distractor 15 wrongly assumes Mira got 5 correct.

Q29. [DILR-2 · mcsc · diff 0.5]

How many questions did Nina answer correctly?

- A) 4 questions
- B) 3 questions ✓
- C) 2 questions
- D) 5 questions

Answer: B) 3 questions

Solution: With Owen = 2 correct, totals summing to 9, the strict order Mira > Nina > Owen, and nobody reaching 5, the only fit is Mira = 4, Nina = 3, Owen = 2. So Nina answered 3 correctly. Distractor 4 = Mira's count; distractor 2 = Owen's count; distractor 5 is impossible since no one got all 5.

Q30. [DILR-2 · mcsc · diff 0.5]

What was the combined total score of all three friends?

- A) 27 points ✓
- B) 30 points
- C) 9 points
- D) 18 points

Answer: A) 27 points

Solution: The three answered 9 questions correctly in total, each worth 3 points, so the combined score is $3 \times 9 = 27$ points (Mira 12 + Nina 9 + Owen 6 = 27). Distractor 9 = total correct answers, not points; distractor 18 = scoring only 2 points per correct; distractor 30 misadds the individual scores.

Instructions (DILR-3): study the shared data below and answer its 4 question(s).

Units sold during the month (number of packets)

Product	Aria	Bani	Cera	Dovi
Tea	120	200	150	180
Coffee	80	150	200	120
Cocoa	60	100	50	90

Selling price per packet (in rupees)

Product	Price
Tea	20
Coffee	30
Cocoa	40

Q31. [DILR-3 · mcsc · diff 0.5]

How many packets in total did store Bani sell across all three products during the month?

- A) 350
- B) 400
- C) 450 ✓
- D) 260

Answer: C) 450

Solution: Store Bani's column: Tea 200 + Coffee 150 + Cocoa 100 = 450 packets. Distractor 400 is Cera's column total, 260 is Aria's column total, and 350 drops the cocoa figure.

Q32. [DILR-3 · mcsc · diff 0.5]

What was the total sales revenue (in rupees) earned by store Cera from all three products?

- A) 12500
- B) 11000 ✓
- C) 10800
- D) 7200

Answer: B) 11000

Solution: Cera: Tea $150 \times 20 = 3000$, Coffee $200 \times 30 = 6000$, Cocoa $50 \times 40 = 2000$; total $3000 + 6000 + 2000 = 11000$. Distractor 12500 is Bani's revenue, 10800 is Dovi's, and 7200 is Aria's.

Q33. [DILR-3 · mcsc · diff 0.5]

What was the combined revenue (in rupees) from Tea across all four stores?

- A) 650
- B) 16500
- C) 11000
- D) 13000 ✓

Answer: D) 13000

Solution: Tea packets sold: $120+200+150+180 = 650$; at 20 rupees each the revenue is $650 \times 20 = 13000$. Distractor 650 forgets to multiply by price, 16500 multiplies by coffee's price 30 (wrong rate), and 11000 is Cera's total revenue.

Q34. [DILR-3 · mcsc · diff 0.5]

In how many of the four stores did the total number of packets sold (all products combined) exceed 350?

- A) 4
- B) 3 ✓
- C) 1
- D) 2

Answer: B) 3

Solution: Store totals are Aria 260, Bani 450, Cera 400, Dovi 390. Three of them (Bani, Cera, Dovi) exceed 350; only Aria does not. Distractor 2 miscounts Dovi, 4 wrongly includes Aria, and 1 counts only Bani.

Instructions (DILR-4): study the shared data below and answer its 3 question(s).

Three investors, Priya, Rohan and Sara, each put a whole number of lakhs of rupees into a separate fund at the start of the year. The following facts are known:

- Together the three invested a total of 24 lakh. - Rohan invested exactly twice as much as Priya. - Sara invested 4 lakh more than Priya.

Over the year the funds gave these returns on the amount invested: Priya's fund returned 20%, Rohan's fund returned 10%, and Sara's fund returned 30%.

Q35. [DILR-4 · mcsc · diff 0.5]

How much did Rohan invest?

- A) 5 lakh
- B) 10 lakh ✓
- C) 9 lakh
- D) 12 lakh

Answer: B) 10 lakh

Solution: Let Priya invest P . Then Rohan = $2P$ and Sara = $P + 4$. The total gives $P + 2P + (P + 4) = 24$, so $4P + 4 = 24$, $P = 5$. Hence Rohan = $2 \times 5 = 10$ lakh. Distractor 5 = Priya's amount; distractor 9 = Sara's amount; distractor 12 solves the equation incorrectly.

Q36. [DILR-4 · mcsc · diff 0.5]

What was the combined amount invested by Priya and Sara together?

- A) 19 lakh
- B) 10 lakh
- C) 15 lakh
- D) 14 lakh ✓

Answer: D) 14 lakh

Solution: From $P + 2P + (P + 4) = 24$ we get $P = 5$, so Priya = 5 and Sara = $5 + 4 = 9$. Together they invested $5 + 9 = 14$ lakh. Distractor 10 = Rohan's amount; distractor 15 = Priya plus Rohan; distractor 19 = Rohan plus Sara.

Q37. [DILR-4 · mcsc · diff 1]

After the year's returns are added to the amounts invested, which investor ends with the largest total amount of money?

- A) Rohan and Sara end level, each with 11 lakh
- B) Sara, with 11.7 lakh ✓
- C) Rohan, with 11 lakh
- D) Priya, with 6 lakh

Answer: B) Sara, with 11.7 lakh

Solution: Investments: Priya 5, Rohan 10, Sara 9 lakh. Adding returns: Priya = $5 \times 1.20 = 6.0$; Rohan = $10 \times 1.10 = 11.0$; Sara = $9 \times 1.30 = 11.7$. The largest final amount is Sara's 11.7 lakh, just above Rohan's 11.0. Distractor 2 stops at Rohan, ignoring Sara's higher return rate; distractor 3 is the smallest total; distractor 4 wrongly treats them as tied (11.7 not equal to 11.0).

Instructions (DILR-5): study the shared data below and answer its 5 question(s).

Units produced by a factory in each quarter

Product	Q1	Q2	Q3
Steel	400	500	600
Cement	300	300	450
Glass	200	250	200
Wood	100	150	250

Production cost per unit (in rupees)

Product	Rate
Steel	50
Cement	40
Glass	60
Wood	30

Q38. [DILR-5 · mcsc · diff 0.5]

What was the total number of Steel units produced across the three quarters?

- A) 1500 ✓
- B) 1050
- C) 1100
- D) 1300

Answer: A) 1500

Solution: Steel row: $400 + 500 + 600 = 1500$ units. Distractor 1050 is Cement's three-quarter total, 1300 sums all four products in Q1+Q2 only, and 1100 drops one quarter.

Q39. [DILR-5 · mcsc · diff 0.5]

By what percentage did Steel production increase from Q1 to Q3?

- A) 20
- B) 200
- C) 33
- D) 50 ✓

Answer: D) 50

Solution: Steel rose from 400 in Q1 to 600 in Q3, an increase of 200; $200/400 \times 100 = 50$ percent. Distractor 20 uses the Q1-to-Q2 step over the wrong base, 200 reports the raw increase as a percentage, and 33 computes the increase over Q3 instead of Q1.

Q40. [DILR-5 · mcsc · diff 0.5]

What was the average quarterly production of Cement over the three quarters?

- A) 300
- B) 1050
- C) 350 ✓
- D) 375

Answer: C) 350

Solution: Cement: $(300 + 300 + 450)/3 = 1050/3 = 350$ units. Distractor 300 takes only the most frequent value, 1050 is the three-quarter sum (not the average), and 375 averages the wrong figures.

Q41. [DILR-5 · mcsc · diff 0.5]

What is the largest single value of units produced by any one product in any one quarter?

- A) 500
- B) 450
- C) 250
- D) 600 ✓

Answer: D) 600

Solution: Scanning every cell, the maximum is Steel in Q3 with 600 units. Distractor 500 is Steel in Q2, 450 is the top Cement figure, and 250 is the highest Wood figure.

Q42. [DILR-5 · mcsc · diff 0.5]

What was the total production cost (in rupees) of all four products in Q1?

- A) 47000 ✓
- B) 44000
- C) 1000
- D) 50000

Answer: A) 47000

Solution: Q1 costs: Steel $400 \times 50 = 20000$, Cement $300 \times 40 = 12000$, Glass $200 \times 60 = 12000$, Wood $100 \times 30 = 3000$; total $20000 + 12000 + 12000 + 3000 = 47000$. Distractor 1000 sums only the Q1 units, 44000 omits the Wood term, and 50000 applies a flat rate of 50 to every product.

Section: QA (suggested 40 min)

Q43. [QA-MCQ-1 · mcsc · diff 0]

A shopkeeper buys an article for 80 euros and sells it for 100 euros. What is the profit percentage?

- A) 20%
- B) 80%
- C) 125%
- D) 25% ✓

Answer: D) 25%

Solution: Profit = $100 - 80 = 20$; profit% = $20/80 \times 100 = 25\%$. The 20% value wrongly uses the selling price as the base (20/100). The 80% value just restates the cost figure as a percentage. The 125% value is the SP-to-CP ratio, not the profit rate.

Q44. [QA-MCQ-2 · mcsc · diff 0]

The ratio of boys to girls in a class is 3 : 5. If there are 20 girls, how many boys are there?

- A) 33 boys
- B) 15 boys
- C) 8 boys
- D) 12 boys ✓

Answer: D) 12 boys

Solution: One part = $20/5 = 4$ students; boys = $3 \times 4 = 12$. The 33 value sets girls equal to 5 parts of value 5 plus 20 wrongly. The 15 value takes 3/4 of 20 (treats total as 4 parts). The 8 value takes 2 parts, using a 3:5 difference incorrectly.

Q45. [QA-MCQ-3 · mcsc · diff 0]

What is the average of the five numbers 4, 8, 10, 12 and 16?

- A) 10 ✓
- B) 50
- C) 12
- D) 8

Answer: A) 10

Solution: Sum = $4+8+10+12+16 = 50$; average = $50/5 = 10$. The 50 value gives the sum without dividing by 5. The 12 value is the median-like middle pick, not the mean. The 8 value divides the sum by 6 instead of 5 (off-by-one count error).

Q46. [QA-MCQ-4 · mcsc · diff 0]

A circle has radius 7 cm. What is its area? (use $\pi = 22/7$)

- A) 44 cm^2
- B) 154 cm^2 ✓
- C) 616 cm^2
- D) 49 cm^2

Answer: B) 154 cm^2

Solution: Area = $\pi r^2 = (22/7)(7^2) = (22/7)(49) = 154 \text{ cm}^2$. The 44 value is the circumference $2\pi r$, not the area. The 616 value uses diameter 14 as the radius ($\pi(14^2)$). The 49 value gives r^2 alone, forgetting the factor π .

Q47. [QA-MCQ-5 · mcsc · diff 0]

Solve for x: $3x + 7 = 22$.

- A) $29/3$
- B) 9
- C) 5 ✓
- D) 15

Answer: C) 5

Solution: $3x = 22 - 7 = 15$, so $x = 15/3 = 5$. The $29/3$ value adds 7 instead of subtracting it ($22+7 = 29$). The 9 value subtracts 7 from 22 but then subtracts 3 instead of dividing. The 15 value stops at $3x = 15$ without dividing by 3.

Q48. [QA-MCQ-6 · mcsc · diff 0]

A car travels 150 km in 3 hours. What is its average speed?

- A) 50 km/h ✓
- B) 450 km/h
- C) 75 km/h
- D) 2 km/h

Answer: A) 50 km/h

Solution: Speed = distance/time = $150/3 = 50 \text{ km/h}$. The 450 value multiplies distance by time. The 75 value divides by 2 instead of 3. The 2 value computes time/distance ($3/150$ scaled), inverting the formula.

Q49. [QA-MCQ-7 · mcsc · diff 0.5]

An article marked at 500 euros is sold after a 20% discount, and the seller still makes a 25% profit on cost. What was the cost price?

- A) 400 euros
- B) 333 euros
- C) 320 euros ✓
- D) 300 euros

Answer: C) 320 euros

Solution: Selling price = $500 \times 0.80 = 400$. With 25% profit, $SP = 1.25 \times CP$, so $CP = 400/1.25 = 320$. The 300 value divides 400 by 1.2 (uses the discount rate as the profit rate). The 400 value reports the selling price, not the cost. The 333 value takes $400/1.2$ rounded, again misusing the rate.

Q50. [QA-MCQ-8 · mcsc · diff 0.5]

A sum of 8000 euros is invested at 10% per annum compound interest. What is the amount after 2 years?

- A) 1680 euros
- B) 9700 euros
- C) 9600 euros
- D) 9680 euros ✓

Answer: D) 9680 euros

Solution: Amount = $8000 \times (1.10)^2 = 8000 \times 1.21 = 9680$. The 9600 value uses simple interest ($8000 + 2 \times 800$). The 1680 value gives only the compound interest, not the total amount. The 9700 value adds an extra rounding bump, miscomputing 1.21.

Q51. [QA-MCQ-9 · mcsc · diff 0.5]

Pipe A fills a tank in 12 hours and pipe B fills it in 6 hours. If both pipes are opened together, how long do they take to fill the tank?

- A) 4 hours ✓
- B) 9 hours
- C) 18 hours
- D) 3 hours

Answer: A) 4 hours

Solution: Combined rate = $1/12 + 1/6 = 1/12 + 2/12 = 3/12 = 1/4$ tank/hour, so time = 4 hours. The 9 value averages the two times. The 18 value adds the two times. The 3 value takes half of the faster pipe's time arbitrarily.

Q52. [QA-MCQ-10 · mcsc · diff 0.5]

When a positive integer n is divided by 7 the remainder is 4. What is the remainder when $3n$ is divided by 7?

- A) 4
- B) 12
- C) 2
- D) 5 ✓

Answer: D) 5

Solution: $n = 7k + 4$, so $3n = 21k + 12 = 7(3k+1) + 5$, giving remainder 5. The 4 value keeps the original remainder unchanged. The 12 value gives 3×4 without reducing modulo 7. The 2 value subtracts 7 twice from 12 instead of once.

Q53. [QA-MCQ-11 · mcsc · diff 0.5]

The HCF of two numbers is 6 and their LCM is 180. If one number is 36, what is the other number?

- A) 30 ✓
- B) 180
- C) 1080
- D) 5

Answer: A) 30

Solution: Product of numbers = HCF $\times$ LCM = $6 \times 180 = 1080$; other number = $1080/36 = 30$. The 180 value confuses the LCM with the missing number. The 1080 value gives the product itself, not the second number. The 5 value divides 180 by 36 (uses LCM instead of HCF $\times$ LCM).

Q54. [QA-MCQ-12 · mcsc · diff 0.5]

A bag contains 4 red and 6 blue balls. Two balls are drawn at random without replacement. What is the probability that both are red?

- A) $2/15$ ✓
- B) $4/25$
- C) $1/5$
- D) $2/9$

Answer: A) $2/15$

Solution: $P = (4/10)(3/9) = 12/90 = 2/15$. The $4/25$ value uses $(4/10)^2$, i.e. with replacement. The $1/5$ value is $C(4,2)/C(6,2)$ using the wrong denominator. The $2/9$ value takes $(4/9)(\dots)$ mishandling the first draw count.

Q55. [QA-MCQ-13 · mcsc · diff 0.5]

How many distinct arrangements can be made using all the letters of the word LEVEL?

- A) 120
- B) 30 ✓
- C) 60
- D) 20

Answer: B) 30

Solution: LEVEL has 5 letters with L repeated twice and E repeated twice: $5!/(2!2!) = 120/4 = 30$. The 120 value is 5! ignoring repeats. The 60 value divides by only one 2! (corrects a single repeat). The 20 value divides by 3! erroneously.

Q56. [QA-MCQ-14 · mcsc · diff 0.5]

In a group of 50 people, 30 like tea and 25 like coffee. If 10 like neither, how many like both tea and coffee?

- A) 5
- B) 15 ✓
- C) 55
- D) 40

Answer: B) 15

Solution: People liking at least one = $50 - 10 = 40$. By inclusion-exclusion, both = $30 + 25 - 40 = 15$. The 5 value forgets to subtract the 10 (uses total 50: $30+25-50$). The 55 value just adds $30+25$. The 40 value reports the union, not the intersection.

Q57. [QA-MCQ-15 · mcsc · diff 0.5]

The third term of an arithmetic progression is 11 and the seventh term is 27. What is the first term?

- A) 5
- B) 4
- C) 7
- D) 3 ✓

Answer: D) 3

Solution: $a + 6d = 27$ and $a + 2d = 11$, so $4d = 16$, $d = 4$; then $a = 11 - 2(4) = 3$. The 4 value gives the common difference d . The 7 value uses $a + 4d = 27$ (miscounts the term gap). The 5 value takes $11 - 6$ by guessing $d = 3$.

Q58. [QA-MCQ-16 · mcsc · diff 1]

Two trains start at the same time from stations A and B, 330 km apart, moving towards each other at 60 km/h and 50 km/h. How far from A do they meet?

- A) 180 km ✓
- B) 150 km
- C) 165 km
- D) 110 km

Answer: A) 180 km

Solution: Closing speed = $60 + 50 = 110$ km/h; meeting time = $330/110 = 3$ h; distance from A = $60 \times 3 = 180$ km. The 150 value is the distance the slower train covers (from B). The 165 value is the midpoint, ignoring the speed difference. The 110 value mistakes the closing speed for the distance.

Q59. [QA-MCQ-17 · mcsc · diff 1]

Vessel A holds 40 litres of a mixture with milk and water in the ratio 3 : 1, and vessel B holds 20 litres in the ratio 1 :

1. If both are poured into one vessel, what is the ratio of milk to water in the new mixture?

- A) 2 : 1 ✓
- B) 4 : 3
- C) 5 : 3
- D) 3 : 2

Answer: A) 2 : 1

Solution: A: milk = 30, water = 10. B: milk = 10, water = 10. Total milk = 40, water = 20, ratio $40:20 = 2:1$. The 3:2 value averages the two ratios. The 4:3 value mixes up B's split as 2:1. The 5:3 value errs by using 25 litres of milk from A.

Q60. [QA-MCQ-18 · msc · diff 1]

A rectangular garden is 40 m long and 30 m wide. A path of uniform width 2 m runs all around it on the inside. What is the area of the path?

- A) 280 m²
- B) 144 m²
- C) 264 m² ✓
- D) 536 m²

Answer: C) 264 m²

Solution: Outer area = $40 \times 30 = 1200$. The inner rectangle shrinks by 2 m on each side, so it is $(40 - 4) \times (30 - 4) = 36 \times 26 = 936$. Path area = $1200 - 936 = 264 \text{ m}^2$. The 280 value subtracts width only once per side ($38 \times 28 = 1064$ misstep). The 144 value is just the corner squares plus a slip. The 536 value treats the path as a 2 m strip on the outside of a 40×30 frame, doubling the perimeter contribution.

Q61. [QA-TITA-1 · tita · diff 0]

Evaluate the expression $(2^5 \times 2^3) \div 2^4$. Type your answer as an integer.

Answer (TITA): 16

Solution: Using the laws of indices, the numerator $2^5 \times 2^3 = 2^{(5+3)} = 2^8$. Dividing by 2^4 gives $2^{(8-4)} = 2^4 = 16$. So the value of the expression is 16.

Q62. [QA-TITA-2 · tita · diff 0]

An item is priced at 80 euros. Its price is first increased by 25%, then the new price is decreased by 20%. What is the final price, in euros? Type your answer as an integer.

Answer (TITA): 80

Solution: A 25% increase multiplies the price by 1.25, giving $80 \times 1.25 = 100$. A subsequent 20% decrease multiplies by 0.80, giving $100 \times 0.80 = 80$. The successive +25% then -20% changes cancel exactly here, so the final price is 80 euros.

Q63. [QA-TITA-3 · tita · diff 0.5]

Using the digits 1, 2, 3, 4 and 5 without repetition, how many three-digit numbers divisible by 5 can be formed? Type your answer as an integer.

Answer (TITA): 12

Solution: A number is divisible by 5 only if its last digit is 0 or 5. Since 0 is not available, the units digit must be 5 (1 way). The hundreds digit can then be any of the remaining 4 digits, and the tens digit any of the remaining 3 digits: $4 \times 3 = 12$ numbers. Total = 12.

Q64. [QA-TITA-4 · tita · diff 0.5]

In a class, 30 students study French and 25 study German. Exactly 10 students study both languages. How many students study exactly one of the two languages? Type your answer as an integer.

Answer (TITA): 35

Solution: Students studying only French = $30 - 10 = 20$. Students studying only German = $25 - 10 = 15$. Those studying exactly one language = $20 + 15 = 35$. Equivalently, $|A| + |B| - 2|A \cap B| = 30 + 25 - 20 = 35$.

Q65. [QA-TITA-5 · tita · diff 0.5]

In an arithmetic progression the first term is 7 and the common difference is 5. What is the value of the 12th term? Type your answer as an integer.

Answer (TITA): 62

Solution: The n th term of an arithmetic progression is $a + (n-1)d$. With $a = 7$, $d = 5$ and $n = 12$, the 12th term = $7 + (12-1) \times 5 = 7 + 11 \times 5 = 7 + 55 = 62$.

Q66. [QA-TITA-6 · tita · diff 1]

A vessel contains 64 litres of pure wine. 16 litres are drawn out and replaced with water; this operation (drawing 16 litres of the mixture and replacing it with water) is performed two more times, for a total of three times. How many litres of pure wine remain in the vessel? Type your answer as an integer.

Answer (TITA): 27

Solution: After each replacement of 16 litres out of 64, the fraction of wine remaining is multiplied by $(64-16)/64 = 48/64 = 3/4$. After three operations the wine left = $64 \times (3/4)^3 = 64 \times 27/64 = 27$ litres. So 27 litres of pure wine remain.